```python
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from datetime import datetime, timedelta
import numpy as np

# --- Style ---
plt.rcParams['font.family'] = 'DejaVu Sans'
plt.rcParams['font.size'] = 10
fig, ax = plt.subplots(figsize=(18, 10))
fig.patch.set_facecolor('#0d1117')
ax.set_facecolor('#0d1117')

# --- Approximate QUBT price data (monthly anchored + insider trade prices) ---
price_dates = [
    datetime(2025, 1, 15), datetime(2025, 2, 15), datetime(2025, 3, 15),
    datetime(2025, 3, 25),  # Huang sell
    datetime(2025, 4, 15), datetime(2025, 5, 1),
    datetime(2025, 5, 19),  # Huang + Shabani sell
    datetime(2025, 6, 1),
    datetime(2025, 6, 9),   # Boehmler + Turmelle sell
    datetime(2025, 6, 12),  # Boehmler sell more
    datetime(2025, 7, 1), datetime(2025, 7, 15),
    datetime(2025, 8, 1), datetime(2025, 8, 15),
    datetime(2025, 8, 25),  # Pouya sell
    datetime(2025, 9, 4),   # Huang sell
    datetime(2025, 9, 9),   # Fagenson sell
    datetime(2025, 9, 15),
    datetime(2025, 10, 3),  # 52-week high
    datetime(2025, 10, 15), datetime(2025, 11, 1),
    datetime(2025, 11, 15), datetime(2025, 12, 1),
    datetime(2025, 12, 17), # LSI acquisition announced
    datetime(2026, 1, 7),   # Milan sell
    datetime(2026, 1, 15), datetime(2026, 2, 1),
    datetime(2026, 2, 15),  # LSI acquisition closed
    datetime(2026, 3, 10),  # Roberts sell
    datetime(2026, 3, 30),  # 52-week low
    datetime(2026, 4, 15), datetime(2026, 5, 1),
    datetime(2026, 5, 12),  # Earnings gap up
    datetime(2026, 5, 17),  # Today
]
prices = [
    7.50, 7.80, 8.20,
    8.48,    # Huang
    9.50, 10.80,
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    11.70,   # Huang + Shabani
    13.50,
    15.50,   # Boehmler + Turmelle
    20.00,   # Boehmler
    18.50, 17.00,
    16.50, 16.00,
    15.53,   # Pouya
    14.42,   # Huang
    15.56,   # Fagenson
    17.00,
    25.84,   # 52w high
    22.00, 19.00,
    16.00, 14.00,
    13.00,   # LSI announced
    11.85,   # Milan
    11.00, 10.50,
    10.00,   # LSI closed
    7.85,    # Roberts
    6.18,    # 52w low
    8.00, 9.00,
    14.45,   # Earnings gap
    10.51,   # Today
]

# Plot price line
ax.plot(price_dates, prices, color='#58a6ff', linewidth=2, zorder=3, alpha=0.9)
ax.fill_between(price_dates, prices, alpha=0.05, color='#58a6ff')

# --- Insider sells (red dots) ---
sells = [
    (datetime(2025, 3, 25), 8.48, "Huang\n200K sh\n$1.7M", -80),
    (datetime(2025, 5, 19), 11.70, "Huang 500K + Shabani 40K\n$6.3M", -90),
    (datetime(2025, 6, 9), 15.50, "CFO Boehmler 272K\nDir Turmelle 201K\n$7.4M",
    (datetime(2025, 6, 12), 20.00, "CFO Boehmler 46K\n$929K", -70),
    (datetime(2025, 8, 25), 15.53, "CRO Pouya 17K\n(sold 100%)", -65),
    (datetime(2025, 9, 4), 14.42, "CEO Huang 1M sh\n$14.4M", 40),
    (datetime(2025, 9, 9), 15.56, "Fagenson 100K\n(sold 100%)\n$1.6M", -80),
    (datetime(2026, 1, 7), 11.85, "COO Milan\n3K sh", -50),
    (datetime(2026, 3, 10), 7.85, "CFO Roberts 78K\n$615K", -60),
]

for date, price, label, offset in sells:
    ax.scatter(date, price, color='#ff4444', s=120, zorder=5, edgecolors='white',
    ax.annotate(label, (date, price),
                textcoords="offset points", xytext=(10, offset),
                fontsize=7.5, color='#ff6666',
                arrowprops=dict(arrowstyle='-', color='#ff444466', lw=0.8),
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                bbox=dict(boxstyle='round,pad=0.3', facecolor='#1a1a2e', edgecolo

# --- Corporate events (vertical lines) ---
events = [
    (datetime(2022, 6, 16), "QPhoton\nacquired", '#ffcc00'),
    (datetime(2025, 12, 17), "Luminar Semi\nacquisition\nannounced\n$110M", '#00c
    (datetime(2026, 2, 1), "Luminar Semi\nclosed", '#00cc88'),
    (datetime(2026, 3, 31), "NuCrypt\nacquired", '#00cc88'),
]

for date, label, color in events:
    if date >= datetime(2025, 1, 1):
        ax.axvline(x=date, color=color, linestyle='--', alpha=0.4, linewidth=1, z
        ax.annotate(label, (date, max(prices) * 0.95),
                    fontsize=7.5, color=color, ha='center', va='top',
                    bbox=dict(boxstyle='round,pad=0.3', facecolor='#0d1117', edge

# --- Insider exit reference zone ---
exit_low = 8.48
exit_high = 20.00
exit_core_low = 11.66
exit_core_high = 15.56

ax.axhspan(exit_core_low, exit_core_high, alpha=0.08, color='#ff4444', zorder=1)
ax.annotate('INSIDER EXIT ZONE\n$11.66 – $15.56\n(core selling range)',
            xy=(datetime(2025, 1, 20), (exit_core_low + exit_core_high) / 2),
            fontsize=8, color='#ff6666', alpha=0.7, va='center',
            bbox=dict(boxstyle='round,pad=0.3', facecolor='#1a1a2e', edgecolor='#

# --- CLUSTER SELL highlight ---
cluster_start = datetime(2025, 9, 1)
cluster_end = datetime(2025, 9, 12)
ax.axvspan(cluster_start, cluster_end, alpha=0.15, color='#ff4444', zorder=1)
ax.annotate('CLUSTER SELL\n3 insiders\n10 days\n$16.5M',
            xy=(datetime(2025, 9, 6), 22),
            fontsize=8, color='#ff4444', ha='center', fontweight='bold',
            bbox=dict(boxstyle='round,pad=0.4', facecolor='#1a1a2e', edgecolor='#

# --- 52-week high annotation ---
ax.annotate('52W HIGH\n$25.84', (datetime(2025, 10, 3), 25.84),
            textcoords="offset points", xytext=(15, 10),
            fontsize=8, color='#ffcc00',
            arrowprops=dict(arrowstyle='->', color='#ffcc00', lw=1),
            bbox=dict(boxstyle='round,pad=0.3', facecolor='#1a1a2e', edgecolor='#

# --- Key narrative note ---
ax.annotate(
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    'NARRATIVE ARC:\nInsiders sold $33M+\nbetween $8-$20\nbefore stock peaked at
    xy=(datetime(2026, 4, 15), 3),
    fontsize=9, color='#aaaaaa', ha='center',
    bbox=dict(boxstyle='round,pad=0.5', facecolor='#1a1a2e', edgecolor='#333333')

# --- Formatting ---
ax.set_xlim(datetime(2025, 1, 1), datetime(2026, 5, 20))
ax.set_ylim(0, 30)
ax.xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
ax.xaxis.set_major_locator(mdates.MonthLocator(interval=2))
ax.yaxis.set_major_formatter('${x:.0f}')
ax.tick_params(colors='#888888')
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.spines['bottom'].set_color('#333333')
ax.spines['left'].set_color('#333333')
ax.grid(axis='y', color='#222222', linewidth=0.5)

# Title
ax.set_title('QUBT — Insider Sell Timeline vs. Stock Price\nQuantum Computing Inc
             fontsize=16, color='#e6e6e6', fontweight='bold', pad=20)
ax.set_ylabel('Stock Price (USD)', color='#888888', fontsize=11)

# Legend
from matplotlib.lines import Line2D
legend_elements = [
    Line2D([0], [0], color='#58a6ff', lw=2, label='Stock Price'),
    Line2D([0], [0], marker='o', color='w', markerfacecolor='#ff4444', markersize
    Line2D([0], [0], color='#00cc88', lw=1, linestyle='--', label='Corporate Even
    Line2D([0], [0], color='#ff4444', lw=6, alpha=0.15, label='Cluster Sell Windo
]
ax.legend(handles=legend_elements, loc='upper left', fontsize=9,
          facecolor='#0d1117', edgecolor='#333333', labelcolor='#aaaaaa')

# Company history subtitle
fig.text(0.5, 0.01,
         'Ticketcart (2001, ink cartridges) → Innovative Beverage (2007) → Ceased
         ha='center', fontsize=8, color='#555555', style='italic')

plt.tight_layout()
plt.savefig('/home/claude/qubt-case/charts/QUBT_timeline.png', dpi=200, bbox_inch
            facecolor='#0d1117', edgecolor='none')
print("Chart saved.")
```